

5.5.2 Electromagnetic radiation from stars

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) energy levels of electrons in isolated gas atoms	THIS WILL ALSO REQUIRE KNOWLEDGE OF 4.5
(b) the idea that energy levels have negative values	
(c) emission spectral lines from hot gases in terms of emission of photons and transition of electrons between discrete energy levels	
(d) the equations $hf = \Delta E$ and $\frac{hc}{\lambda} = \Delta E$	
(e) different atoms have different spectral lines which can be used to identify elements within stars	

- (f) continuous spectrum, emission line spectrum and absorption line spectrum
- (g) transmission diffraction grating used to determine the wavelength of light
- (h) the condition for maxima $d\sin\theta = n\lambda$, where d is the grating spacing
- (i) use of Wien's displacement law $\lambda_{\max} \propto \frac{1}{T}$ to estimate the peak surface temperature (of a star)
- (j) luminosity L of a star;
Stefan's law $L = 4\pi r^2 \sigma T^4$
where σ is the Stefan constant
- (k) use of Wien's displacement law and Stefan's law to estimate the radius of a star.

The structure and use of an optical spectrometer are not required;

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Proof of this equation is not required.

**THIS WILL ALSO REQUIRE
KNOWLEDGE OF 4.4.1**

